

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is the server could not handle the request coming repeatedly from an unfamiliar IP address.

The logs show that the server is flooded with SYN packets.

This event could be a SYN flood attack.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The SYN packets are the initial request from the client to the server sending a synchronized message to receive information or connect with the server. This is the first step required to make a TCP connection. SYN stands for "synchronize".
2. The SYN-ACK packets is the response from the web server agreeing for the connection. The server will reserve system resources for the final step of the handshake. SYN, ACK stands for "synchronize acknowledge."
3. The ACK packet is the client acknowledging the permission to connect. This is the final step required to make a successful TCP connection. ACK stands for "acknowledge."

When a malicious actor sends a large number of SYN packets all at once to the server, the server gets overwhelmed and is unable to respond with the requests.

The logs indicate that a user with an unfamiliar IP address is trying to establish connection with the web server, but instead flooded the server with the SYN packets. Before, employees could access the Web server but after multiple incoming requests, Server was declining requests from employees and further was not sending the SYN-ACK packets to even the malicious actor, resulting in the server to shut down, or failing to respond.